

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE

$$S_n = n \left[\frac{a_1 + a_2}{2} \right]$$

Exercise

Sequence & Series

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. If $a_1, a_2, a_3, \dots, a_n, \dots$ are in A.P. such that $a_4 - a_7 + a_{10} = m$, then the sum of first 13 terms of this A.P., is :
 (A) 15 m (B) 10 m (C) 12 m (D) 13 m SS0001
2. If a, b, c are in AP, then $(a - c)^2$ equals
 (A) $4(b^2 - ac)$ (B) $4(b^2 + ac)$ (C) $4b^2 - ac$ (D) $b^2 - 4ac$ SS0002
3. Let $a_1, a_2, a_3, \dots$ be an A.P. such that $\frac{a_1 + a_2 + \dots + a_p}{a_1 + a_2 + a_3 + \dots + a_q} = \frac{p^2}{q^2}$; $p \neq q$. Then $\frac{a_6}{a_{21}}$ is equal to :
 (A) $\frac{121}{1861}$ (B) $\frac{11}{41}$ (C) $\frac{121}{1681}$ (D) $\frac{41}{11}$ SS0003
4. Given sum of the first n terms of an A. P. is $2n + 3n^2$. Another A. P. is formed with the same first term and double of the common difference, the sum of n terms of the new A. P. is :-
 (A) $n + 4n^2$ (B) $n^2 + 4n$ (C) $3n + 2n^2$ (D) $6n^2 - n$ SS0004
5. If for an A.P. $a_1, a_2, a_3, \dots, a_n, \dots$
 $a_1 + a_3 + a_5 = -12$ and $a_1 a_2 a_3 = 8$, then the value of $a_2 + a_4 + a_6$ equals
 (A) -12 (B) -16 (C) -18 (D) -21 SS0021
6. If the sum of the first 11 terms of an arithmetical progression equals that of the first 19 terms, then the sum of its first 30 terms, is
 (A) equal to 0 (B) equal to -1 (C) equal to 1 (D) non unique SS0022
7. Let $s_1, s_2, s_3, \dots$ and $t_1, t_2, t_3, \dots$ are two arithmetic sequences such that $s_1 = t_1 \neq 0$; $s_2 = 2t_2$ and
 $\sum_{i=1}^{10} s_i = \sum_{i=1}^{15} t_i$. Then the value of $\frac{s_2 - s_1}{t_2 - t_1}$ is
 (A) $8/3$ (B) $3/2$ (C) $19/8$ (D) 2 SS0023
8. Along a road lies an odd number of stones placed at intervals of 10 m. These stones have to be assembled around the middle stone. A person can carry only one stone at a time. A man carried out the job starting with the stone in the middle, carrying stones in succession, thereby covering a distance of 4.8 km. Then the number of stones is
 (A) 15 (B) 29 (C) 31 (D) 35 SS0025

9. In an A.P. with first term 'a' and the common difference d ($a, d \neq 0$), the ratio ' ρ ' of the sum of the first n terms to sum of n terms succeeding them does not depend on n. Then the ratio $\frac{a}{d}$ and the ratio ' ρ ', respectively are

(A) $\frac{1}{2}, \frac{1}{4}$ (B) $2, \frac{1}{3}$ (C) $\frac{1}{2}, \frac{1}{3}$ (D) $\frac{1}{2}, 2$

SS0026

10. Let $a_n, n \in \mathbb{N}$ is an A.P. with common difference 'd' and all whose terms are non-zero. If n approaches infinity, then the sum $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_n a_{n+1}}$ will approach

(A) $\frac{1}{a_1 d}$ (B) $\frac{2}{a_1 d}$ (C) $\frac{1}{2a_1 d}$ (D) $a_1 d$

SS0027

11. The first term of an infinite G.P. is 1 and every term is equal to the sum of the successive terms, then its fourth term will be-

(A) $\frac{1}{2}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{16}$

SS0006

12. If $a = \sum_{n=0}^{\infty} x^n, b = \sum_{n=0}^{\infty} y^n, c = \sum_{n=0}^{\infty} (xy)^n$ where $|x|, |y| < 1$; then-

(A) $abc = a + b + c$ (B) $ab + bc = ac + b$ (C) $ac + bc = ab + c$ (D) $ab + ac = bc + a$

SS0007

13. If $r > 1$ and $x = a + \frac{a}{r} + \frac{a}{r^2} + \dots$ to ∞ , $y = b - \frac{b}{r} + \frac{b}{r^2} - \dots$ to ∞ and $z = c + \frac{c}{r^2} + \frac{c}{r^4} + \dots$ to ∞ , then

$\frac{xy}{z} =$

(A) $\frac{ab}{c}$ (B) $\frac{ac}{b}$ (C) $\frac{bc}{a}$ (D) None of these

SS0008

14. In a GP, first term is 1. If $4T_2 + 5T_3$ is minimum, then its common ratio is

(A) $\frac{2}{5}$ (B) $-\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $-\frac{3}{5}$

SS0009

15. Given a sequence of 4 numbers, first three of which are in G.P. and the last three are in A.P. with common difference six. If first and last terms of this sequence are equal, then the last term is :
 (A) 8 (B) 16 (C) 2 (D) 4

SS0010

16. If G be the GM between x and y, then the value of $\frac{1}{G^2 - x^2} + \frac{1}{G^2 - y^2}$ is equal to
 (A) G^2 (B) $\frac{2}{G^2}$ (C) $\frac{1}{G^2}$ (D) $3G^2$

SS0011

17. If $\frac{1+3+5+\dots\text{upto } n \text{ terms}}{4+7+10+\dots\text{upto } n \text{ terms}} = \frac{20}{7 \log_{10} x}$ and

$$n = \log_{10} x + \log_{10} x^{\frac{1}{2}} + \log_{10} x^{\frac{1}{4}} + \log_{10} x^{\frac{1}{8}} + \dots + \infty, \text{ then } x \text{ is equal to}$$

- (A) 10^3 (B) 10^5 (C) 10^6 (D) 10^7

SS0029

18. If $a \neq 1$ and $\ln a^2 + (\ln a^2)^2 + (\ln a^2)^3 + \dots = 3 \left(\ln a + (\ln a)^2 + (\ln a)^3 + (\ln a)^4 + \dots \right)$ then 'a' is equal to
 (A) $e^{1/5}$ (B) $\sqrt{e}$ (C) $\sqrt[3]{e}$ (D) $\sqrt[4]{e}$

SS0030

19. If a, b, c are in HP, then $\frac{a-b}{b-c}$ is equal to

- (A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) $\frac{a}{c}$ (D) $\frac{c}{b}$

SS0012

20. If a, b, c are distinct positive real in H.P., then the value of the expression, $\frac{b+a}{b-a} + \frac{b+c}{b-c}$ is equal to
 (A) 1 (B) 2 (C) 3 (D) 4

SS0031

21. An H.M. is inserted between the number $1/3$ and an unknown number. If we diminish the reciprocal of the inserted number by 6, it is the G.M. of the reciprocal of $1/3$ and that of the unknown number. If all the terms of the respective H.P. are distinct then
 (A) the unknown number is 27 (B) the unknown number is $1/27$
 (C) the H.M. is 15 (D) the G.M. is 21

SS0032

22. If $x = \sum_{n=0}^{\infty} a^n$, $y = \sum_{n=0}^{\infty} b^n$, $z = \sum_{n=0}^{\infty} c^n$ where a, b, c are in A.P. and $|a| < 1$, $|b| < 1$, $|c| < 1$, then x, y, z are in-
 (A) HP (B) Arithmetic - Geometric Progression
 (C) AP (D) GP

SS0035

23. If a, b and c are positive real numbers then $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is greater than or equal to
 (A) 3 (B) 6 (C) 27 (D) 5
 SS0013
24. If $a_1, a_2, a_3, \dots, a_n \in \mathbb{R}^+$ and $a_1 \cdot a_2 \cdot a_3 \dots a_n = 1$, then minimum value of $(1 + a_1 + a_1^2)(1 + a_2 + a_2^2) \dots (1 + a_n + a_n^2)$ is equal to :-
 (A) 3^{n+1} (B) 3^n (C) 3^{n-1} (D) none of these
 SS0014
25. If a, b, c are positive real numbers such that $ab^2c^3 = 64$ then minimum value of $\left(\frac{1}{a} + \frac{2}{b} + \frac{3}{c}\right)$ is equal to:-
 (A) 6 (B) 2 (C) 3 (D) None of these
 SS0015
26. If $abcd = 1$ where a, b, c, d are positive reals then the minimum value of $a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd$ is
 (A) 6 (B) 10 (C) 12 (D) 20
 SS0033
27. If $x \in \mathbb{R}$, the numbers $(5^{1+x} + 5^{1-x}), a/2, (25^x + 25^{-x})$ form an A.P. then 'a' must lie in the interval
 (A) $[1, 5]$ (B) $[2, 5]$ (C) $[5, 12]$ (D) $[12, \infty)$
 SS0024
28. $2 + 4 + 7 + 11 + 16 + \dots$ to n terms =
 (A) $\frac{1}{6}(n^2 + 3n + 8)$ (B) $\frac{n}{6}(n^2 + 3n + 8)$ (C) $\frac{1}{6}(n^2 - 3n + 8)$ (D) $\frac{n}{6}(n^2 - 3n + 8)$
 SS0018
29. The sum $\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \dots$ upto 11-terms is :-
 (A) $\frac{11}{4}$ (B) $\frac{60}{11}$ (C) $\frac{7}{2}$ (D) $\frac{11}{2}$
 SS0019
30. The sum of the series : $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots$ up to 10 terms, is:
 (A) $\frac{22}{13}$ (B) $\frac{18}{11}$ (C) $\frac{20}{11}$ (D) $\frac{16}{9}$
 SS0020

EXERCISE (O-2)

Multiple Correct Answer Type

1. If $a_1, a_2, \dots, a_n$ are distinct terms of an A.P., then
 (A) $a_1 + 2a_2 + a_3 = 0$ (B) $a_1 - 2a_2 + a_3 = 0$
 (C) $a_1 + 3a_2 - 3a_3 - a_4 = 0$ (D) $a_1 - 4a_2 + 6a_3 - 4a_4 + a_5 = 0$ **SS0126**
2. Let $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ be arithmetic progressions such that $a_1 = 25, b_1 = 75$ and $a_{100} + b_{100} = 100$. Then
 (A) the difference between successive terms in progression 'a' is opposite of the difference in progression 'b'.
 (B) $a_n + b_n = 100$ for any n .
 (C) $(a_1 + b_1), (a_2 + b_2), (a_3 + b_3), \dots$ are in A.P.
 (D) $\sum_{r=1}^{100} (a_r + b_r) = 10000$ **SS0038**
3. If $1, \log_y x, \log_z y, -15 \log_x z$ are in A.P., then
 (A) $z^3 = x$ (B) $x = y^{-1}$ (C) $z^{-3} = y$ (D) $x = y^{-1} = z^3$ **SS0127**
4. Real number x_1, x_2, x_3 in A.P. satisfying the equation $x^3 - x^2 + \alpha x + \beta = 0$ then select correct alternative(s)
 (A) $\alpha \leq \frac{1}{3}$ (B) $\beta \geq -\frac{1}{27}$ (C) $\alpha > \frac{1}{3}$ (D) $\beta < -\frac{1}{27}$ **SS0128**
5. Let a, b, c are non zero real numbers such that $3(a^2 + b^2 + c^2 + 1) = 2(a + b + c + ab + bc + ca)$ then
 (A) a, b, c are in AP (B) a, b, c are in GP
 (C) a, b, c are in HP (D) a^2, b^2, c^2 are in GP **SS0129**
6. If a satisfies the equation $a^{2017} - 2a + 1 = 0$ and $S = 1 + a + a^2 + \dots + a^{2016}$ then possible value(s) of S is/are
 (A) 2016 (B) 2018 (C) 2017 (D) 2 **SS0130**
7. If $a_1, a_2, a_3, \dots$ are positive numbers in G.P., then which of the following is/are true -
 (A) $a_1^3, a_2^3, a_3^3, \dots$ are in G.P. (B) $\frac{a_2}{a_1}, \left(\frac{a_3}{a_2}\right)^2, \left(\frac{a_4}{a_3}\right)^3, \dots$ are in G.P.
 (C) $\ln a_1, \ln a_2, \ln a_3, \dots$ are in A.P. (D) $a_1 a_2, a_2 a_3, a_3 a_4, \dots$ are in G.P. **SS0131**
8. The increasing geometric sequence of natural numbers $x_1, x_2, x_3, \dots$ consist entirely of integral powers of 2. Given $\sum_{n=1}^8 \log_2(x_n) = 196$ then which of the following is/are correct?
 (A) Total number of all possible such sequences is 4
 (B) Total number of all possible such sequences is more than four
 (C) Maximum value of x_4 is 2^{24}
 (D) Maximum value of x_4 is 2^{23} **SS0132**

9. If equation $16x^4 - mx^3 + (2m + 17)x^2 - mx + 16 = 0$ has four distinct roots forming a geometric progression, and one of the root is $\frac{1}{2}$, then :

(A) common ratio of G.P. is 4

(B) $m = 170$

(C) sum of the roots = $\frac{85}{8}$

(D) Product of root = 3

SS0133

10. Three numbers a, b, c between 2 and 18 are such that

(i) their sum is 25

(ii) the numbers 2, a, b , are in A.P.

(iii) the number $b, c, 18$ are in G.P.

then which of the following options are correct.

(A) $a = 5$

(B) $b = 8$

(C) $b + c = 20$

(D) $a + b + c = 25$ SS0134

11. Let $\{a_n\}$ be a sequence such that $a_0 = 1, a_1 = 0, a_n = 3a_{n-1} - 2a_{n-2}$. The correct statement is: SS0135

(A) $\sum_{i=1}^{50} a_i = 2(51 - 2^{50})$ (B) $a_{51} = 2^{25} - 2$ (C) $a_{48} = 2(1 - 2^{47})$ (D) $\sum_{i=1}^{49} a_i = (2^{49} - 50)$

12. If a, b, c are positive integers forming first three terms of increasing geometric progression with common ratio 'r', where $b - a$ is perfect cube and $\log_6 a + \log_6 b + \log_6 c = 6$. Let S denotes sum of 10 terms of another G.P. whose first term is 'a' and common ratio is $\frac{5r}{2}$, then

(A) $a + b + c = 189$

(B) $a + b + c = 273$

(C) sum of digits of S is 90.

(D) $S = 10^{10} - 1$

SS0136

13. The roots of the equation $x^4 - 8x^3 + ax^2 - bx + 16 = 0$, are positive, if

(A) $a = 24$

(B) $a = 12$

(C) $b = 8$

(D) $b = 32$

SS0137

14. Choose the correct statement(s)

(A) Sum of ten arithmetic means between $\sin^2 18^\circ$ and $\cos^2 18^\circ$ is 5

(B) Product of ten geometric means between $4\tan 15^\circ$ and $\cot 15^\circ$ is 1024

(C) If $27abc \geq (a + b + c)^3$ and $3a + 4b + 5c = 12$ then $\frac{1}{a^2} + \frac{1}{b^3} + \frac{1}{c^5} = 10$; where a, b, c are positive real numbers.

(D) For positive real numbers A.M. $\geq$ G.M.

SS0138

15. Let $x, y, z \in \mathbb{R}^+$ and $x + y + z = 2$, then choose the correct option(s)-

(A) maximum value of $x^2 y^3 z^2$ is $\frac{2^{11} \cdot 3^3}{7^7}$

(B) maximum value of $x^2 y^3 z^2$ is 1.

(C) minimum value of $\frac{4}{x} + \frac{1}{y} + \frac{9}{z}$ is 18

(D) maximum value of xyz is $\frac{8}{27}$

SS0139

16. Let $0 < a < b$ such that $2, 2^x - \frac{7}{2}, 2^x - 5$ are harmonic mean, arithmetic mean and geometric mean of a, b respectively then which of the following option(s) is/are correct
 (A) number of possible values of x is 2 (B) number of value of x is 1
 (C) $a^2 + b^2$ can be 63 (D) $a^2 + b^2$ can be 3 **SS0140**

17. The value of $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{5^n}$ is $\frac{a}{b}$, where a & b are relatively prime then
 (A) $a > b$ (B) $a < b$ (C) $a > 3$ (D) $b > 10$ **SS0141**

18. Let $S_r = 2 + 4 + 6 + \dots + r$ terms then -

- (A) $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{S_r} = 1$ (B) $\sum_{r=1}^n S_r = \frac{n(n+1)(n+2)}{6}$
 (C) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sum_{r=1}^k S_r} = \frac{3}{4}$ (D) $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sum_{r=1}^k S_r} = \frac{1}{4}$ **SS0142**

19. The value of $\sum_{r=2}^{100} \frac{2^r(2-r)}{r(r+1)(r+2)}$ is equal to

- (A) $\frac{2}{3} - \frac{2^{100}}{(51)(101)}$ (B) $\frac{2}{3} - \frac{2^{101}}{(101)(102)}$ (C) $\frac{1}{2} - \frac{2^{100}}{100 \cdot 101}$ (D) $\frac{1}{2} - \frac{2^{100}}{(101)(51)}$ **SS0143**

20. Consider the arrangement

1
 2 3 4
 5 6 7 8 9
 10 11 12 13 14 15 16

and so on

then correct option's is/are

- (A) first term of 15th row is 197
 (B) first term of 15th row is 201

- (C) let t_i be the number of terms in i^{th} row then $\sum_{i=1}^n t_i = n^2$

- (D) let t_i be the number of terms in i^{th} row then $\sum_{i=1}^n t_i = 2n - 1$ **SS0144**

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 2

A geometric series has first term 'a' and common ratio 'r'. The second term of the series is 4 and the sum upto infinite terms of the series is 25

1. The sum of all possible values of a is-
(A) 5 (B) 15 (C) 20 (D) 25 **SS0145**
2. If 'r' is taken the larger of its two possible values then the smallest value of n for which S_n exceeds 15 is-
(A) 4 (B) 5 (C) 6 (D) 7 **SS0146**

Paragraph for Question 3 to 4

From first 'n' natural numbers, 2 consecutive numbers are removed the arithmetic mean of the remaining number is $\frac{105}{4}$ then answer the following

3. n is divisible by
(A) 15 (B) 25 (C) 5 (D) 10 **SS0147**
4. The removed number is(are)
(A) 6 (B) 7 (C) 8 (D) 9 **SS0148**

Paragraph for Question 5 to 6

In a G.P. the sum of the first and last term is 66 and the product of second and the second last term is 128. Sum of all terms is 126

On the basis of above information, answer the following questions :

5. If an increasing G.P. is considered then number of terms **SS0149**
6. Difference of greatest and least term is **SS0150**

Matrix Match Type

7. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

- (P) If p, q, r are in GP ; $a^p = b^q = c^r$, then ℓna , ℓnb , ℓnc are always in
- (Q) If $\frac{a+be^x}{a-be^x} = \frac{b+ce^x}{b-ce^x} = \frac{c+de^x}{c-de^x}$, then a, b, c, d are in
- (R) If a, b, c are in AP; a, p, b are in GP; b, q, c are in GP, then p^2 , b^2 , q^2 are in
- (S) If a, b, c are in GP, then $\log_a 10$, $\log_b 10$, $\log_c 10$ are in

List-II

- (1) AP
- (2) GP
- (3) HP
- (4) None

Codes :

- | | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 1 | 4 | 3 | 1 |
| (B) | 4 | 3 | 1 | 2 |
| (C) | 3 | 4 | 2 | 2 |
| (D) | 2 | 2 | 1 | 3 |

SS0151

8. Match List-I with List-II and select the correct answer using the code given below the list.

List-I**List-II**

- (P) Number of common integers in two arithmetic progression 1, 3, 5, 7, ... 157 and 2, 5, 8, 11 ... 161 is (1) 25
- (Q) If a and b are roots of $11x^2 - 4x - 2 = 0$, then value of $10(1 + a + a^2 + \dots) (1 + b + b^2 + \dots)$ is (2) 26
- (R) If the sum of first 20 terms is same as the sum of first 10 terms in an arithmetic progression then the sum of first 30 terms is less than (3) 22

- (S) If sum of $\frac{3}{1^2 \cdot 2^2} + \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} + \dots + \frac{199}{99^2 \cdot 100^2}$ is $\frac{a}{b}$ (4) 20
(where a and b are coprime) the value of $20|b - a|$ is (5) 10

- (A) P $\rightarrow$ 2; Q $\rightarrow$ 1; R $\rightarrow$ 1, 2, 3; S $\rightarrow$ 3 (B) P $\rightarrow$ 2; Q $\rightarrow$ 3; R $\rightarrow$ 1, 2, 3, 4, 5; S $\rightarrow$ 4
(C) P $\rightarrow$ 1; Q $\rightarrow$ 2; R $\rightarrow$ 1, 2; S $\rightarrow$ 2 (D) P $\rightarrow$ 1; Q $\rightarrow$ 4; R $\rightarrow$ 3, 4; S $\rightarrow$ 1

SS0152

Match the column

9. **Column - I**

Column - II

- (A) The coefficient of x^{49} in the product $(x - 1)(x - 3)(x - 5)(x - 7) \dots (x - 99)$ (p) -2500
- (B) Let S_n denote sum of first n terms of an A.P. If $S_{2n} = 3S_n$, then $\frac{S_{3n}}{S_n}$ is (q) 9

- (C) The sum $\sum_{r=2}^{\infty} \frac{1}{r^2 - 1}$ is equal to (r) 3/4

- (D) The length, breadth, height of a rectangular box are in G.P. (length > breadth > height) The volume is 27, the total surface area is 78. Then the length is (s) 6

SS0153

10. **Column - I**

Column - II

- (A) The value of xyz is 15/2 or 18/5 according as the series a, x, y, z, b are in an A.P. or H.P. then 'a + b' equals where a, b are positive integers. (p) 2
- (B) The value of $2^{\frac{1}{4}} 4^{\frac{1}{8}} 8^{\frac{1}{16}} \dots \infty$ is equal to (q) 1
- (C) If x, y, z are in A.P., then $(x + 2y - z)(2y + z - x)(z + x - y) = kxyz$, where $k \in \mathbb{N}$, then k is equal to (r) 3
- (D) There are m A.M. between 1 and 31. If the ratio of the (s) 4

7^{th} and $(m - 1)^{\text{th}}$ means is 5 : 9, then $\frac{m}{7}$ is equal to

SS0154

EXERCISE (O-4)

Numerical Grid Type

1. The interior angles of a convex polygon form an arithmetic progression with a common difference of 4° . Determine the number of sides of the polygon if its largest interior angle is 172° .

SS0040

2. The sequence $a_1, a_2, a_3, \dots, a_{98}$ satisfies the relation $a_{n+1} = a_n + 1$ for $n = 1, 2, 3, \dots, 97$ and has the sum equal to 4949. Evaluate $\sum_{k=1}^{49} a_{2k}$.

SS0042

3. The first term of an arithmetic progression is 1 and the sum of the first nine terms equal to 369. The first and the ninth term of a geometric progression with real common ratio coincide with the first and the ninth term of the arithmetic progression. Find the seventh term of the geometric progression.

SS0044

4. For an increasing G.P. $a_1, a_2, a_3, \dots, a_n$, if $a_6 = 4a_4$, $a_9 - a_7 = 192$, then the value of $\sum_{i=1}^{\infty} \frac{1}{a_i}$ is

SS0045

5. If $a > 0$, then minimum value of $a + 2a^2 + a^3 + 15 + a^{-1} + a^{-3} + a^{-4}$ is

SS0041

6. If $a, b, c, d > 0$ such that $a + 2b + 3c + 4d = 50$, then find the maximum value of $\left(\frac{a^2 b^4 c^3 d}{16} \right)^{1/10}$

SS0051

7. If $x > 0$, then find greatest value of the expression $\frac{x^{100}}{1 + x + x^2 + x^3 + \dots + x^{200}}$.

SS0155

8. If number of coins earned in n^{th} game is $n2^{n+2} - 2^n$ and total number of coins earned in first 10 games is $10(B \cdot 2^{10} + 1)$, where $B \in \mathbb{N}$, then the value of B is

SS0052

9. Find the sum of the sequence $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \dots \infty$

SS0067

10. Let $S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}}$. Find $[S]$.

Where $[y]$ denotes largest integer less than or equal to y .

SS0069

EXERCISE (JM)

1. If m is the A.M. of two distinct real numbers l and n ($l, n > 1$) and G_1, G_2 and G_3 are three geometric means between l and n , then $G_1^4 + 2G_2^4 + G_3^4$ equals - [JEE(Main)-2015]
 (1) $4lmn^2$ (2) $4l^2m^2n^2$ (3) $4l^2mn$ (4) $4lm^2n$ SS0070
2. If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is:- [JEE(Main)-2016]
 (1) $\frac{7}{4}$ (2) $\frac{8}{5}$ (3) $\frac{4}{3}$ (4) 1 SS0071
3. If the sum of the first ten terms of the series $\left(1\frac{3}{5}\right)^2 + \left(2\frac{2}{5}\right)^2 + \left(3\frac{1}{5}\right)^2 + 4^2 + \left(4\frac{4}{5}\right)^2 + \dots$ is $\frac{16}{5}m$, then m is equal to :- [JEE(Main)-2016]
 (1) 99 (2) 102 (3) 101 (4) 100 SS0072
4. If, for a positive integer n , the quadratic equation, $x(x+1) + (x+1)(x+2) + \dots + (x+n-1)(x+n) = 10n$ has two consecutive integral solutions, then n is equal to : [JEE(Main)-2017]
 (1) 11 (2) 12 (3) 9 (4) 10 SS0073
5. For any three positive real numbers a, b and c , $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$. Then : [JEE(Main)-2017]
 (1) a, b and c are in G.P. (2) b, c and a are in G.P.
 (3) b, c and a are in A.P. (4) a, b and c are in A.P. SS0074
6. Let $a_1, a_2, a_3, \dots, a_{49}$ be in A.P. such that $\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$. If $a_1^2 + a_2^2 + \dots + a_{17}^2 = 140m$, then m is equal to- [JEE(Main)-2018]
 (1) 68 (2) 34 (3) 33 (4) 66 SS0075
7. Let A be the sum of the first 20 terms and B be the sum of the first 40 terms of the series $1^2 + 2 \cdot 2^2 + 3^2 + 2 \cdot 4^2 + 5^2 + 2 \cdot 6^2 + \dots$. If $B - 2A = 100\lambda$, then λ is equal to : [JEE(Main)-2018]
 (1) 248 (2) 464 (3) 496 (4) 232 SS0076
8. If a, b and c be three distinct real numbers in G. P. and $a + b + c = xb$, then x cannot be : [JEE(Main)-2019]
 (1) 4 (2) -3 (3) -2 (4) 2 SS0077

9. Let $a_1, a_2, \dots, a_{30}$ be an A. P., $S = \sum_{i=1}^{30} a_i$ and $T = \sum_{i=1}^{15} a_{(2i-1)}$. If $a_5 = 27$ and $S - 2T = 75$, then a_{10} is equal to : [JEE(Main)-2019]
- (1) 57 (2) 47 (3) 42 (4) 52

SS0078

10. The sum of the following series

$$1 + 6 + \frac{9(1^2 + 2^2 + 3^2)}{7} + \frac{12(1^2 + 2^2 + 3^2 + 4^2)}{9} + \frac{15(1^2 + 2^2 + \dots + 5^2)}{11} + \dots \text{ up to 15 terms, is :}$$

[JEE(Main)-2019]

- (1) 7820 (2) 7830
(3) 7520 (4) 7510

SS0079

11. Let x, y be positive real numbers and m, n positive integers. The maximum value of the expression

$$\frac{x^m y^n}{(1 + x^{2m})(1 + y^{2n})} \text{ is :-}$$

[JEE(Main)-2019]

- (1) $\frac{1}{2}$ (2) $\frac{1}{4}$
(3) $\frac{m+n}{6mn}$ (4) 1

SS0080

12. If α, β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to :

[JEE(Main)-2019]

- (1) $\beta\gamma$ (2) 0 (3) $\alpha\gamma$ (4) $\alpha\beta$

SS0081

13. The sum of all natural numbers 'n' such that $100 < n < 200$ and H.C.F. $(91, n) > 1$ is :

[JEE(Main)- 2019]

- (1) 3221 (2) 3121 (3) 3203 (4) 3303

SS0082

14. If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct?

[JEE(Main)-2019]

- (1) d, e, f are in A.P. (2) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.
(3) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P. (4) d, e, f are in G.P.

SS0083

15. Let the sum of the first n terms of a non-constant A.P., $a_1, a_2, a_3, \dots$ be $50n + \frac{n(n-7)}{2}A$, where A is a constant. If d is the common difference of this A.P., then the ordered pair (d, a_{50}) is equal to
[JEE(Main)- 2019]

(1) $(A, 50+46A)$ (2) $(A, 50+45A)$ (3) $(50, 50+46A)$ (4) $(50, 50+45A)$

SS0084

16. The sum $\frac{3 \times 1^3}{1^2} + \frac{5 \times (1^3 + 2^3)}{1^2 + 2^2} + \frac{7 \times (1^3 + 2^3 + 3^3)}{1^2 + 2^2 + 3^2} + \dots$ upto 10th term, is : [JEE(Main)- 2019]

(1) 660 (2) 620 (3) 680 (4) 600

SS0085

17. If $a_1, a_2, a_3, \dots, a_n$ are in A.P. and $a_1 + a_4 + a_7 + \dots + a_{16} = 114$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to : [JEE(Main)- 2019]

(1) 38 (2) 98 (3) 76 (4) 64

SS0086

18. Let a, b and c be in G. P. with common ratio r , where $a \neq 0$ and $0 < r \leq \frac{1}{2}$. If $3a, 7b$ and $15c$ are the first three terms of an A. P., then the 4th term of this A. P. is : [JEE(Main)- 2019]

(1) $\frac{7}{3}a$ (2) a (3) $\frac{2}{3}a$ (4) $5a$

SS0087

19. If α and β are the roots of the equation $375x^2 - 25x - 2 = 0$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \alpha^r + \lim_{n \rightarrow \infty} \sum_{r=1}^n \beta^r$ is equal to : [JEE(Main)- 2019]

(1) $\frac{21}{346}$ (2) $\frac{29}{358}$ (3) $\frac{1}{12}$ (4) $\frac{7}{116}$

SS0088

20. If the sum of the first 40 terms of the series, $3 + 4 + 8 + 9 + 13 + 14 + 18 + 19 + \dots$ is $(102)m$, then m is equal to : [JEE(Main)- 2020]

(1) 20 (2) 5 (3) 10 (4) 25

SS0089

21. Let $a_1, a_2, a_3, \dots$ be a G.P. such that $a_1 < 0$, $a_1 + a_2 = 4$ and $a_3 + a_4 = 16$. If $\sum_{i=1}^9 a_i = 4\lambda$, then λ is equal to : [JEE(Main)- 2020]

(1) -171 (2) 171 (3) $\frac{511}{3}$ (4) -513

SS0090

22. Five numbers are in A.P., whose sum is 25 and product is 2520. If one of these five numbers is $-\frac{1}{2}$, then the greatest number amongst them is : [JEE(Main)- 2020]

(1) $\frac{21}{2}$ (2) 27 (3) 16 (4) 7

SS0091

23. The greatest positive integer k, for which $49^k + 1$ is a factor of the sum $49^{125} + 49^{124} + \dots + 49^2 + 49 + 1$, is : [JEE(Main)- 2020]

(1) 32 (2) 60 (3) 63 (4) 65

SS0092

24. The sum, $\sum_{n=1}^7 \frac{n(n+1)(2n+1)}{4}$ is equal to _____. [JEE(Main)- 2020]

SS0093

25. The sum $\sum_{k=1}^{20} (1+2+3+\dots+k)$ is [JEE(Main)- 2020]

SS0094

26. The number of terms common to the two A.P.'s 3, 7, 11,, 407 and 2, 9, 16,, 709 is _____. [JEE(Main)- 2020]

SS0095

27. The product $2^{\frac{1}{4}} \cdot 4^{\frac{1}{16}} \cdot 8^{\frac{1}{48}} \cdot 16^{\frac{1}{128}} \cdot \dots$ to ∞ is equal to : [JEE(Main)- 2020]

(1) $2^{\frac{1}{2}}$ (2) $2^{\frac{1}{4}}$ (3) 2 (4) 1

SS0096

28. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_1 = 1$, $a_2 = 1$ and $a_{n+2} = 2a_{n+1} + a_n$ for all $n \geq 1$. Then the value of

$47 \sum_{n=1}^{\infty} \frac{a_n}{2^{3n}}$ is equal to _____. [JEE(Main)- 2021]

SS0156

EXERCISE (JA)

1. If the sum of first n terms of an A.P. is cn^2 , then the sum of squares of these n terms is

(A) $\frac{n(4n^2-1)c^2}{6}$ (B) $\frac{n(4n^2+1)c^2}{3}$ (C) $\frac{n(4n^2-1)c^2}{3}$ (D) $\frac{n(4n^2+1)c^2}{6}$

[JEE 2009, 3 (-1)]

SS0097

2. Let $a_1, a_2, a_3, \dots, a_{11}$ be real numbers satisfying $a_1 = 15$, $27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4, \dots, 11$.

If $\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + \dots + a_{11}}{11}$ is equal to [JEE 2010, 3+3]

SS0098

3. The minimum value of the sum of real numbers a^{-5} , a^{-4} , $3a^{-3}$, 1 , a^8 and a^{10} with $a > 0$ is

[JEE 2011, 4]

SS0099

4. Let $a_1, a_2, a_3, \dots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^p a_i$, $1 \leq p \leq 100$. For any

integer n with $1 \leq n \leq 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n , then a_2 is [JEE 2011, 4]

SS0100

5. Let $a_1, a_2, a_3, \dots$ be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is [JEE 2012, 3 (-1)]

(A) 22 (B) 23 (C) 24 (D) 25

SS0101

6. Let $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$. Then S_n can take value(s) [JEE-Advanced 2013, 4, (-1)]

(A) 1056 (B) 1088 (C) 1120 (D) 1332

SS0102

7. A pack contains n cards numbered from 1 to n . Two consecutive numbered cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller to the numbers on the removed cards is k , then $k - 20 =$ [JEE-Advanced 2013, 4, (-1)]

SS0103

8. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is $b + 2$, then the value of $\frac{a^2 + a - 14}{a + 1}$ is [JEE(Advanced)-2014, 3]

SS0104

9. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is 6 : 11 and the seventh term lies in between 130 and 140, then the common difference of this A.P. is **[JEE 2015, 4M, -0M]**
SS0105
10. Let $b_i > 1$ for $i = 1, 2, \dots, 101$. Suppose $\log_e b_1, \log_e b_2, \dots, \log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, \dots, a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If $t = b_1 + b_2 + \dots + b_{51}$ and $s = a_1 + a_2 + \dots + a_{51}$ then **[JEE(Advanced)-2016, 3(-1)]**
SS0106
 (A) $s > t$ and $a_{101} > b_{101}$ (B) $s > t$ and $a_{101} < b_{101}$
 (C) $s < t$ and $a_{101} > b_{101}$ (D) $s < t$ and $a_{101} < b_{101}$
11. The sides of the right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side ? **[JEE(Advanced)-2017, 3]**
SS0107
12. Let X be the set consisting of the first 2018 terms of the arithmetic progression 1, 6, 11,, and Y be the set consisting of the first 2018 terms of the arithmetic progression 9, 16, 23,, Then, the number of elements in the set $X \cup Y$ is _____ **[JEE(Advanced)-2018, 3]**
SS0108
13. Let m be the minimum possible value of $\log_3 (3^{y_1} + 3^{y_2} + 3^{y_3})$, where y_1, y_2, y_3 are real numbers for which $y_1 + y_2 + y_3 = 9$. Let M be the maximum possible value of $(\log_3 x_1 + \log_3 x_2 + \log_3 x_3)$, where x_1, x_2, x_3 are positive real numbers for which $x_1 + x_2 + x_3 = 9$. Then the value of $\log_2 (m^3) + \log_3 (M^2)$ is _____. **[JEE(Advanced)-2020]**
SS0124
14. Let $a_1, a_2, a_3, \dots$ be a sequence of positive integers in arithmetic progression with common difference 2. Also, let $b_1, b_2, b_3, \dots$ be a sequence of positive integers in geometric progression with common ratio 2. If $a_1 = b_1 = c$, then the number of all possible values of c, for which the equality

$$2(a_1 + a_2 + \dots + a_n) = b_1 + b_2 + \dots + b_n$$
 holds for some positive integer n, is _____ **[JEE(Advanced)-2020]**
SS0125
15. Let $\ell_1, \ell_2, \dots, \ell_{100}$ be consecutive terms of an arithmetic progression with common difference d_1 , and let $w_1, w_2, \dots, w_{100}$ be consecutive terms of another arithmetic progression with common difference d_2 , where $d_1 d_2 = 10$. For each $i = 1, 2, \dots, 100$, let R_i be a rectangle with length ℓ_i , width w_i and area A_i . If $A_{51} - A_{50} = 1000$, then the value of $A_{100} - A_{90}$ is _____. **SS0157**
[JEE(Advanced)-2022]

ANSWER KEY

Do yourself - 1 :

1. (a) $\frac{2}{1}, \frac{4}{2}, \frac{8}{3}, \frac{16}{4}, \dots$, (b) $\frac{2}{3}, \frac{4}{9}, \frac{2}{27}, \frac{4}{81}, \dots$;
 3. $-(p+q)$ 4. 153 5. 0 6. -185
 7. $820a - 1680b$

Do yourself - 2 :

1. 900 2. 14 3. 3 4. 5 5. 612
 6. 4, 9, 14 7. 1, 4, 7 8. 495 9. n^2 10. 3, 5, 7, 9
 11. $6r - 1$ 13. $(14n - 6)/(8n + 23)$ 14. $2Q$

Do yourself - 3 :

1. 931 2. 4^5 4. 4, 12, 36 5. (C) 6. $191\frac{1}{4}$
 7. $\frac{1}{4}(5^p - 1)$ 8. $364(\sqrt{3} + 1)$ 9. $\frac{64}{65}$ 10. $\frac{27}{58}$
 11. $\frac{3(3 + \sqrt{3})}{2}$ 12. 2 13. 16, 24, 36, ... 14. 8, 12, 18
 15. 2, 6, 18 16. $6, -3, 1\frac{1}{2}, \dots$ 17. $4, 1, \frac{1}{4}, \dots$
 18. $\frac{x(x^n - 1)}{x - 1} + \frac{(n(n+1))a}{2}$ 19. $n = 38$

Do yourself - 4 :

1. 2 2. 15 4. (a) 5 (b) 6 (c) -18
 8. 1 9. (iii) $b = 4, c = 6, d = 9$ or $b = -2, c = -6, d = -18$

Do yourself - 5 :

1. $\frac{1}{2}$ 3. $-\frac{1}{4}, -\frac{3}{4}, \dots, -9\frac{1}{4}$ 4. $1, -1\frac{1}{2}, \dots, -39$.
 5. $\frac{3}{2}, 1, \frac{2}{3}$ 6. $\frac{16}{3}, 8, \dots, 27$ 7. $6\frac{1}{9}, 7\frac{6}{7}$
 8. $\frac{2}{5}, \frac{2}{7}, \frac{2}{9}, \frac{2}{11}$ 10. (a) 6, 3 (b) 120, 30 12. 28 13. 102

Do yourself - 6 :

1. $4 + \frac{8}{9} \left(1 - \frac{1}{4^{n-1}} \right) - \left(\frac{2n+1}{3 \times 4^{n-1}} \right)$
2. $\frac{1-a^n}{(1-a)^2} - \frac{na^n}{1-a}$
3. $\frac{8}{3}$
4. $\frac{9}{8}$
5. $n \cdot 2^{n+2} - 2^{n+1} + 2$
6. 23

Do yourself - 7 :

1. $\frac{n(n+3)}{4}$
2. (a) $\frac{n(n+1)(2n+3)}{2}$ (b) $n^2(n+1)$
- (c) $\frac{1}{4}n(n+1)(n^2+n+3)$ (d) $\frac{1}{6}n(n+1)(2n+7)$
- (e) $\frac{1}{2}n(n+1)(n^2+3n+1)$ (f) $\frac{1}{2}(3^{n+1}+1)-2^{n+1}$
- (g) $4^{n+1} - 4 - n(n+1)(n^2-n-1)$
3. (a) $\frac{n}{4}(n+1)(n+6)(n+7)$ (b) $\frac{n}{4}(n+1)(n+8)(n+9)$
4. (a) $3n^2 + n; n(n+1)^2$
- (b) $3 \cdot 2^n + n + 2; 6(2^n - 1) + \frac{n(n+5)}{2}$
5. (i) $2^{n+1} - 3; 2^{n+2} - 4 - 3n$ (ii) $n^2 + 4n + 1; (1/6)n(n+1)(2n+13) + n$
6. (i) $s_n = (1/24) - [1/\{6(3n+1)(3n+4)\}] ; s_\infty = 1/24$
- (ii) $(1/5)n(n+1)(n+2)(n+3)(n+4)$
- (iii) $n/(2n+1)$
7. C
8. B
9. C

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	B	D	D	A	C	C	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	A	B	A	C	B	D	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	A	B	C	B	D	B	D	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,D	A,B,C,D	A,B,C,D	A,B	A,B,C,D	C,D	A,B,C,D	A,C	A,B,C	A,B,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C	A,C,D	A,D	A,B,D	A,C,D	B,C	B,C,D	A,C	A,B	A,C

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8		
Ans.	D	B	B,C,D	B,C	6.00	62.00	D	B		
Que.	Q.9	A	B	C	D	Q.10	A	B	C	D
Ans.		p	s	r	q		s	p	s	p

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	12	2499	27	2	22	5	$\frac{1}{201}$	7	0.50	49

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	3	1	3	2	1	4	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	2	3	1	1	3	2	3	1
Que.	21	22	23	24	25	26	27	28		
Ans.	1	3	3	504	1540.00	14	1	7		

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	0	8	9 or 3	D	A,D	5	4	9	B
Que.	11	12	13	14	15					
Ans.	6	3748	8.00	1.00	18900.00					